

Thinking about Research Project

- Pig Atlas, does innovation in science, and thus education of that innovation require the creation of new art? Dependence on science education on art??
 - Knowledge creation
 - How photographs and diagrams work together
 - Define terms in ways that are most helpful to self
 - Could just be about knowledge
 - **What do i mean about art or aesthetic form**
 - **Translucent paper**
 - Big hook
- Project idea, **art is the only way , and creation of new art is the only way in which humans can expand their own objectivity, physical creation, pushing our imaginations is how we fill the gaps in science of what we cannot do, perceive the world in a non human way**

MEETING CHECKPOINT 1

Does innovation in science require the creation of new forms of art? – is terrific. To recap our conversation on Saturday, tomorrow afternoon (around 4) we'll be hoping to see some movement on these research questions:

1. What kinds of case studies will you be looking at? Will you look at one case study or a few?
 - a. Translucent paper book
 - b. Pig Atlas
 - c. The science of discovery itself - <https://news.northeastern.edu/2019/11/13/a-cosmic-data-visualization-celebrates-science-journal-natures-150-year-anniversary/>
2. What kinds of resources (archival, published papers, etc.) were you considering using?
3. Are you interested in “art” per se, or aesthetics more broadly?
 - a. I feel like im interested in novel cross sections between art and science
 - i. The atlas feels novel because not only was it created through a likely handmade arduous process of carving but also of printing cross sections.....
 - ii. Its use (flip) was considered in art, and i guess i think that educational science materials can be recategorized as art created with understanding in mind, it's not really left up to interpretation
4. Do you have a particular span of time you'd want to look at? What are the advantages/disadvantages of casting a wide versus a narrow net?

- a. I guess I'd want some examples of times when art became harder to make or things became harder to understand
 - i. **As science becomes less visible,** Understanding becomes a harder ordeal
 - ii.

In the Library!!!

The Pig Atlas: by Erling S Hegre, PHD at Medical college of virginia, professor of anatomy, richmond

- By the department of Visual education in **1967**, printed by Roger Brown in Richmond virginia
- "It is almost fitting that I acknowledge my indebtedness to Mr. John Conway, artist, who prepared all the layout and registrations for this volume. Without his patience, dedication and skill it could not have been completed"
- "Dedicated to H. Hudnall Ware Jr, physician administrator educator and humanitarian with the hope that it will help to speed the training of others who will join him in working toward the total betterment of our society
- This is a printed copy of a reconstruction made several years ago, was printed bc of its popularity
- Reference points throughout (features of embryo on page 11
- Prepared by the Block Surface method, hegre
- <https://anatomypubs.onlinelibrary.wiley.com/doi/abs/10.1002/ar.1090970103>
- 545 levels at intervals of 16 micra
- 16x magnification of the pig
 - Really helping students to visualize this, and also wants to show them that this is art (not to scale)

Maybe instead the question is, or the argument is, to what extent are we confined in science to the bounds and forms of artistic expression, is advancement in art is necessary for the advancement and accessibility of science?

- makes me think about data visualization, is it about displaying the most information or is education about presenting information

Art is an abstraction, and we only abstract when we understand the whole?>>>

Definition of art???

Human application of creative skill? Evoke emotion or feeling? In this case.... Can that be understanding or relatability?

Offloading a personal perception into an external and perceptual-to-other form of art

- And what is education

Pointilism: 1886

- Hamburger's drawings... 1940s

<https://www.drawforbiodiversity.com/en/dessin-scientifique>

"The visual education department, organized in 1947, offers services in support of patient care, teaching, and research. The centrally-organized department offers graphics services to the entire University. It operates a campus-wide television system in support of instruction and has general responsibility for the student self-study center (media library) in Sanger Hall. In addition to its technical support facilities, the department offers professional consultation services to the faculty in the planning and execution of teaching exercise"

<https://theconversation.com/seeing-what-the-naked-eye-cant-4-essential-reads-on-how-scientists-bring-the-microscopic-world-into-plain-sight-211666#:~:text=Though%20hand%20drawings%20have%20since,microscope%20as%20a%20scientist%20would.>

Scientists use art techniques for recording mechanisms, but difference between record and display

I think what I'm trying to say is that for science to be educational, we need art, because you need to be able to express, in the most digestible way possible, what may have taken years to discover and interpret.

Daston and Gallison

- uses scientific atlas's

Three types of objectivity

Bichat hated microscopes

Corners of aesthetic innovation that don't follow these patterns of scientific representation in the Daston and Gallison book

Scientific innovation Requires leaps of aesthetic innovation - imagination

- chemical symbols and

<https://darchive.mblwhoilib.org/search?query=leuckart>

As Albert Einstein once said “the greatest scientists are artists as well.

EXTRA RESEARCH BY SOURCE:

Background: similarities in art and science

17th Century Microscopy: - microscopes made by eyeglass makers?
<https://gizmodo.com/how-17th-century-artists-helped-make-the-microscopic-wo-1736249872>

- 18th Century Dutch Draper: Anthonie Van Leeuwenhoek built over 500 microscopes and worked closely with contemporary artists to illustrate his findings
 - But Robert Hooke's micrographia also had drawings
 - Van leeu wasn't artistically inclined, needed collaboration

<https://www.smithsonianmag.com/smart-news/see-the-original-17th-century-drawings-of-the-microscopic-world-robert-hooke-discovered-3345281/#:~:text=Before%20photo%20and%20video%2C%20illustrations,to%20share%20what%20they%20saw&text=In%20the%2017th%20century%2C%20microscopes,Charles%20Darwin%20to%20the%20Galapagos.>

- 1665 micrographia, Hooke published
- Conrad marten:
<https://www.artgallery.nsw.gov.au/collection/artists/martens-conrad/>

- An Australian landscape painter, joined the HMS Beagle on a scientific survey expedition as the artist on the ship

Did science influence art:

<https://maverickbeyond.com/2019/02/05/microscopy-in-art/>

- Did the invention of the microscope influence art?
 - Talented people across many subjects- Aristotle, Leonardo da Vinci, Galileo etc
 - Sci art movement
 - Art could be created using scientific tools but scientists were still mainly using art skills and recording mechanisms
 - LOOK INTO SCI ART
 - Bifurcation..... Art influencing science, scientists stopped ... doing art

Science needs art:

<https://www.smithsonianmag.com/blogs/national-museum-of-natural-history/2020/04/15/why-science-needs-art/>

- Smithsonian mag about why science needs art
 - Invertebrate zoology department
 - May be hard to explain the microscopic stinging cells that are needed to explain "how a jellyfish stings" without a clear visual
 - Clay model created of a nematocyst - art for education
 - But also science helps scientists understand the natural world
 - Dr Karen Osborn studies the midwater - oceans largest habitat
 - Preserved jars of midwater animals are weird, muted colors, not as fascinating and beautiful as the researchers find them
 - Scientists learn about photography, about framing,
 - Designing exhibits that showcase their photography and work "Life in one cubic foot" exhibit
 - "Graphs figures and illustrations are all more powerful with a touch of artistry" - paying careful

attention to patterns shapes and spatial relationships in illustration sketch and photography

- When photography fails illustration comes into play
 - Couldn't get the full picture of a cassiosome after microscopy photography methods
- Helps highlight sand aesthetic and beauty
 - Why do people spend their whole lives researching these animals ? well color, vibrancy, liveliness, when capturing these animals and showing them to you in real life isn't possible.... Is where art comes in

Art + Science collaboration for more science:

<https://www.forbes.com/councils/forbestechcouncil/2023/05/12/the-art-of-innovation-how-scientific-research-and-the-arts-can-collaborate-to-generate-innovations/>

- Also talk about that visualization of nature magazine
 - "Scientific research and the arts have been regarded as opposites - empirical evidence vs creative expression", expressing one's emotions, feelings, etc vs expressing something more... objectively, finding out what something really IS vs how it is digested by our brains
 - Both required curiosity and experimentation, and willingness for risk <https://direct.mit.edu/leon/article-abstract/43/1/59/45394/Similarities-and-Contrasts-in-the-Creative?redirectedFrom=fulltext>
- Futuristic Science section of the Virtual Catalog of Contemporary Arts
 - Mona lisa, golden mean/ratio to enhance the aesthetic pleasurability of his work,
 - Using bioluminescent materials in fashion, science influencing art
 - Science fiction imagery and art has inspired technology and science

Digital Scientific Visualization:

<https://theconversation.com/seeing-what-the-naked-eye-cant-4-essential-reads-on-how-scientists-bring-the-microscopic-world-into->

plain-

sight-211666#:~:text=Though%20hand%20drawings%20have%20since,microscope%20as%20a%20scientist%20would.

- Microscopy Unlocking dimensions of the living world for scientists and the general public
- Seeing as Identifying
 - Resolution gaps : analogy, on google maps you can see the city as a whole and individual houses but seeing the neighborhoods within that is a harder task
 - Mosaic
 - Also CRYO -EM in 2017 nobel prize
 - Images at or near atomic resolution and preserve natural shapes
- Seeing as Scoping:
- “Researchers have been charting the brain by stitching together multiple snapshots like a photo mosaic”
 - Collaging?
 - Mosaic - Mesopotamia, colored stones shells and ivory
 - late Middle English: from Old French *musaique*, French *mosaique*, from medieval Latin *mosaicus*, *musaicus*, based on Latin *museum* ‘decoration with small square stones’, perhaps ultimately from Greek *mousa* ‘a muse’.
- Seeing as Appreciating
 - Bio art competitions , viz in classroom and art museum enhance science literacy
 - Being an informed citizen
 - Using art that speaks to people, like music, to give people different intuitions
 - Mapping amino acids into notes and structural loops into tempos and motifs
 - Like how a major and minor chord have an intuition of its darkness or happiness
 - This idea of the artistry of life
 - Beauty of science

<https://theconversation.com/art-illuminates-the-beauty-of-science-and-could-inspire-the-next-generation-of-scientists-young-and-old-168925>

- Explaining the unexplainable
- Where we don't have an accessible nomination of the visible, we should instead, just show the visible
- Flemish Artist Peter Paul Rubens- famous anatomical drawings

- These could then be published in field guides or encyclopedias for further researchers
- Scientific knowledge being shared
 - <https://science.nasa.gov/missions/webb/nasas-webb-takes-star-filled-portrait-of-pillars-of-creation/#:~:text=Webb's%20new%20view%20of%20the,still%20forming%20come%20into%20view.>
 - When a visualization is powerful, it is often revisited, further studied and.... Eventually new discoveries can be uncovered

My own thought- bc visualization makes science more accessible and distributable, it also allows for new perceptions and new viewings likely to occur . Someone else sees something different in the same image

<https://www.faseb.org/awards/bioart>

- Bio art awards
- All about Effective communication
- Jack Kolberg-Edelbrock, super molecular confetti “false-colored”

https://appel.nasa.gov/2012/07/17/47s_art_sci_tech-html/

- At SVS, we use the same computer graphics animation software Pixar employs to create animated films like *Toy Story* and *Wall-E*, but we have added our own software modules to the commercial software to ensure the images we create reflect as accurately as possible the data they are based on.
- The rounds and expertise of SVS members are diverse, but we share a passion for computer graphics, an appreciation or passion for art, and an interest or expertise in science.
- ipal investigator Valerie Casasanto believes that “audiences wont be able to grasp such complex scientific concepts or even pay attention unless they are visually appealing. Through the use of SVS visuals combined with music, an audience can understand a scientific concept without being lectured to. In one look, the whole story is told. This may help inspire young people to study these sciences or take action for better stewardship of our home planet.”
- The NASA Scientific Visualization Studio (SVS) has used RenderMan for its projects. **The studio has also collaborated with RenderMan by providing 3D assets for a RenderMan challenge**
 - **CREATED BY PIXAR!!!!**

Examples of art in science

<https://annemarieatkinson.co.uk/what-is-sciart/>

- SciArt may help us to explore, understand, or express something about the natural world, human psychology, outer-space or more. It may lead to new development in either the field of arts or sciences (or both). It may use creative tools to make science more understandable to the public, or scientific tools to make contemporary art more relatable. It doesn't necessarily need to lead to break-through discoveries or *Eureka!* moments.

Art born out of science:

<https://www.youtube.com/watch?v=99gMbK2QCKE>

- Making data science understandable, communicating what we see in numbers

<https://sc24.supercomputing.org/2024/10/sculpting-data-how-hpc-and-art-create-the-future-of-scientific-visualization/> - advancing visualization, so in the beginning should talk about how science is art (visualization is necessary for communication and discovery is only discovery if it matters to people)

- Cutting Vis Project
 - Consulted artists - samsel to collab on expanding vis language for complex data
 - Line shape form color texture
 - Samsel was a training and practicing and teaching artist before going into scientific viz
 - Co creation
 - Artistic vision vs data complexity
 - Representing highly multivariate data in single images
 - Intuitiveness, accessibility, scientists and non specialists
 - Which..... Back to stippling, multi variate data in a 2d image
 - “We started with glyphs, but we quickly realized that the world of visualization had much more potential,” explained

Samsel. "In the art world, the possibilities for visual encoding, whether through lines, textures, or even movement, are endless. We saw no reason why the same couldn't be applied to scientific data."

- - Expanding our vocabulary, more than the nomination of the visible (parallel growth from natural history.. following art)
 - AR AND VR
 - Bringing data into the world with us rather than a flat screen
 - Bringing out nuanced color schemes
 - Cool glyphs in this article

<https://www.quantamagazine.org/how-ai-revolutionized-protein-science-but-didnt-end-it-20240626/> - alpha fold, importance of visualization ...

- An example of 3d computer graphics
- Innovation in technology (ai) merging with computer graphics
- Pretty accurate and is a diving board for more inquiry, gets you started a lot faster even if not exactly accurate or the same as seeing it with your own eyes

History of Computer Graphics:

<https://www.computerhistory.org/revolution/computer-graphics-music-and-art/15/228>

- Creating art with computers- Bell Labs in the 1960s
- Computer art using punched cards - Shogren at San Jose State University 1963
- Computer art competitions - computers and automations a technical journal
- Galleries began showing computer generated art in the mid 1960s
- MOMA - Suri's Hummingbird 1968
-

<https://unclesp1d3r.github.io/posts/2023/03/computer-history-the-evolution-of-computer-graphics/>

- Computer graphics history
 - 1700s Magic Lantern
 - 1922 Vlad Zworykin CRT electronic tv system
 - Early 1800s. 1832 Charles Babbage analytical engine, calculating and printing math tables, never completed but laid foundation
 - 1960s- first computer based drawing systems, Ivan Sutherland at MIT, SketchPad
 - Mid 1960s, GE developed GE 4D, vector graphics
 - 1970s Picture System by Evans and Sutherland
 - RASTER GRAPHICS: 1970s bitmap display systems
 - The Alto, first Gui
 - 1980s Adobe Illustrator
 - 3d graphics:
 - Rendition Vérité
 - Indigo 2
 - 3dfx Voodoo graphics card
 - 2000s, GPUs, CGI etc, medicine etc

https://en.wikipedia.org/wiki/Maxwell%27s_thermodynamic_surface

- 1874 sculpture representing thermodynamic surfaces. , plaster
- James Clerk Maxwell
- Multivariate - volume, entropy, and energy
 - Putting data on a **convenient** scale for viewing
 - Expressing relationships, not just images

https://en.wikipedia.org/wiki/Computer_graphics#1950s

- Whirlwind, and Sage projects fifth CRT
- One of the first interactive video games to feature recognizable, interactive graphics – *Tennis for Two* – was created for an oscilloscope by William Higinbotham to entertain visitors in 1958 at Brookhaven National Laboratory and simulated a tennis match. In 1959, Douglas T. Ross, while working at MIT on transforming mathematic statements into computer generated 3D mac

https://www.youtube.com/watch?v=hET7R8gJm_c

<https://www.cs.cmu.edu/~ph/nyit/masson/history.htm>

- Early scientific roots, visualizing with construction of physical paper based on mathematical models. - using computation to create a visual representation for science

- 1950s John Whitney Sr. devises his own computer assisted mechanisms to create some of his graphic artwork and short films. One of his sons John Jr. works with and learns from his father from childhood through high school.
-

<https://www.amygoodchild.com/blog/computer-art-50s-and-60s>

- Computing and creativity have always been linked. In the early 1800's when Charles Babbage designed the Analytical Engine, his friend Ada Lovelace wrote in a letter that, if music could be expressed to the engine, then it "might compose elaborate and scientific pieces of music of any degree of complexity or extent." [1]
- mathematician named Ben Laposky was inspired by an article written in Popular Science which suggested that decorative patterns could be created using oscilloscopes. [3]
- In articles and exhibitions of the work at the time, Laposky also included demonstrative examples of the most simple oscillators - perhaps the earliest instance of an artist working with technology finding a way to give their audience more understanding of the work.
- as in Gordon Pask's *MusiColor* (1953). *MusiColor* was a mechanical and electronic system designed to be used by musicians. Sounds from their instruments were input via microphone and analysed for frequency, attack and rhythm. This input determined the output from various lights and pattern or colour wheels controlled by servo motors, with mappings that developed over time. If the input became repetitive, the system would adjust its output to generate more variety in the visual patterns.
 - oundings. It was "difficult or impossible to make genuine use of the system" and *MusiColor* was shelved in 1957. [7]
- 1963 Ken Knowlton invented BEFLIX, a programming language which could output raster animated films. The word pixel wasn't yet in common use and a silent film about the process describes drawing a picture with a "mosaic of squares".
- as unusual in this group as she had worked towards art her whole life and did not have a background in science. She gained access to a computer by going to the Paris University computing centre and explaining to the head

of the department that she wanted to use a one to make art. After giving her a look which implied to Molnar that “he was considering whether he should call for a nurse to sedate me”, he yes anyway. [15]

- **Manfred Mohr: Evolving traditional art**

- Manfred Mohr was a saxophonist and painter and then in 1970 he had the opportunity to work with a Control Data 6400 computer and Benson plotter at the Météorologie Nationale.
- It's clear to see how his artistic and musical career shaped his algorithmic work. His early computer art was “influenced by atonal music, modern Jazz and abstract expressionism.” [18]. A

- **Frieder Nake: Art Advances**

- In 1971, computer artist Frieder Nake wrote a piece titled, perhaps surprisingly, “There should be no computer art”, in which he considers the debate about bringing computers into art from more angles. [20]
- He argues that computer art has *not* contributed “to the advancement of art, if we judge ‘advancement’ by comparing computer products to all existing works for art.” [20] However, on the other hand, he had no doubt that computer art has found “interesting new methods” and that there will be “an entirely new relationship between the creator and the creation.” Nake focuses on the place for computers in the process, rather than the outcome.

- want "computer art" to be a thing in and of itself. The attention should not be on the fact the output was created by a computer, as if that automatically gives it a value but rather that we should seek to make work with value and we can use a computer as a tool to aid us in that.
- Georg Nees first studied mathematics and physics and worked as a mathematician for Siemens, before studying philosophy under Max Bense and publishing his thesis, 'Generative Computergraphik' in 1969.

AI generated answer : The Role of Randomness and Programming Errors: The article emphasizes the conceptual importance of **randomness and chance** in early computer art. It discusses how artists like Vera Molnar used algorithms to explore possibilities and how Georg Nees even incorporated "a fairly serious programming error" into his work. This demonstrates that for these artists, the computer wasn't just a tool for precise control but also a collaborator that could introduce unexpected and interesting results, reflecting a deliberate artistic choice rather than a technical failing.

<https://www.isaloureiro.com/en/2023/06/23/scientific-illustration-vs-photography/>

- Illustration clarifies and simplifies
- Allows you to see inside, non invasively
- Archetype vs individual specimen/holotype
- No model needed? Detailed description, evidence, studies, photos
- Makes invisible visible
- Aesthetic and artistic value - beauty and value of science
 - Aesthetics are important to humans, and scientists, many scientists see their work as fascinating or beautiful

<https://fi.edu/en/blog/why-we-need-scientific-illustration>

- In the advent of cameras, why do we still illustrateIn all stages of science communication, from childhood interest, to secondary education, to professionals and research, illustrators step in to translate the information into materials that fit the needs of individuals at every level of understanding. The ability to transform and craft compelling images, charts, and models that provide accurate information is a skill unique to the scientific illustrator.
-
- Skillful fusion of diff creative thinking
 - Quality and detail of photographs can miss certain things expressible by human sensation that is not still life visual

- Good image of seeing inside (facial anatomy)
- Can show processes instead of just states- animation

Anatomical illustration history:

https://collections.library.utoronto.ca/explore/anatomia/about/history_illustration - color to distinguish different parts of the body

- The publication in 1778 of Soemmering's *De basi encephali* [13] marks the beginning of the final phase, which extends to the mid-nineteenth century. During this period collaboration between anatomist and artist is at its closest. Indeed, many anatomists, such as Soemmering, Charles Bell, Camper, Carswell, and MacLise, were also accomplished artists and drew their designs, while John Bell both drew and engraved many of his. Scarpa employed and trained his own artist, Faustino Anderloni, and supervised his work.
- The introduction of colour plates brought enormous benefit to anatomical illustration, especially in certain branches of medicine. It is a fortunate coincidence that around the time that colour illustration was becoming widespread, great progress was being made in dermatology and pathology. In a work that laid the foundation of modern dermatology, Willan chose colour-printed stipple engraved plates to demonstrate various skin disorders in vivid detail, and in so doing produced beautiful illustrations of an unattractive subject [14]. Colour could not only highlight distinct parts of the body; it could also show changes in natural states brought about by morbid conditions. In their pathology textbooks Carswell [15], and Bright [16] used colour plates to great effect, and contributed greatly to knowledge of morbid anatomy.

Rossi papers; PS. I think I owe you some other readings, too. I've attached one by Michael Lynch about diagrams and photographs; one by Ursula Klein about Berzelian diagrams; and one by Daston and Peter Galison about images and objectivity (there's a lot more Daston and Galison if you're interested). Also, I'm not sure if it's relevant, but I'm putting an article on painting dead fish on the Canvas site, since it relates to our Harvard trip. Look for Aranowski_Deadfish_2016.

Stippling:

- First used in the 15th century as stipple engraving, this technique of dot making combined with line engravings and then only using dots
- Refine the origin: You mention stipple engraving in the 15th century, but a more accurate starting point is the 16th century with Giulio Campagnola. Campagnola used a burin to create dots for smoother tonal transitions in printmaking, distinguishing his work from cross-hatching.
- Photo : <https://risdmuseum.org/art-design/collection/christ-and-woman-samaria-200913>
- <https://www.arthistoryproject.com/artists/giulio-campagnola/>
- Then in the 17th and 18th century someone there was a new thread of stippling that was created.... But derived from the crayon manner (an etched stipple technique that came from using goldsmiths marking wheels and tools to create shading etc and eventually to engrave an entire plate
- Then William Wynne took that technique to Britain and published a book that was a collection of prints in imitation of drawings that belonged to him.
 - Stipple engraving formed
 - Clarify the 18th-century development: Your mention of a "new thread of stippling" in the 17th and 18th centuries could be more specific. This period saw the formal development of stipple engraving in France, known as the "crayon manner." The technique was then perfected and popularized in England by engravers like Francesco Bartolozzi and William Wynne Ryland in the 1760s. Your reference to Ryland is good, but mentioning Bartolozzi and the French origin would provide a more complete picture.

So how make the jump from stipple engraving to stippling in scientific illustration

- Obviously it's the same idea but where is the connect to art?
Why did science NEED art here to get to a form of illustration that dominated centuries to come?

Nuance and realism

I dont have a complete history given the time i spent researching, but

Well stipple engraving was often used to mimic or copy a chalk or pastel drawing into a print - well who was the first to use stippling in a scientific illustration?

- Anatomical drawings followed this technique in anatomy and dermatology and pathology arose
 - Medicine
- This was mostly with color
https://collections.library.utoronto.ca/view/anatomia:RBAI067_0021
- History_illustration
- https://collections.library.utoronto.ca/explore/anatomia/about/history_illustration

And slowly bc of its precision, we begin to see drawings and engravings for scientific illustrations - capturing visual detail rather than approximation in image form - more exact representations, realism

- Today we often think of a photo of the most objective representation scientifically. - bc a photo lets us see it for ourselves, or at least how the photographer wants us to see it. So for scientists to be able to convey honestly what they saw, stippling makes sense.

So we get printers and then images after stippling just become....
Well... photos

But then we get a new medium, a technological advancement

- Computers and computer graphics

- "Because dots are the most simple visual element in a picture, they also are ideal for computer visualizations, " said David S. Ebert, an associate professor in Purdue's School of Electrical and Computer Engineering.
 - ['Stippling' speeds 3-D computer imaging](#)
 - Allowing us to have three dimensional coordinates, x, y, and z (or in the case of stippling, depth of your engraving or pressure of your pencil or paintbrush)

Since then, we've completely advanced our visualization techniques on the computers, we have 3d renderings, movable graphs etc, but your computer is 2d.... And each pixel on your screen is essentially a dot. And the color and brightness or lightness of that dot represents what you see on your laptop

- And even with photos; why did illustration stay around?
- <https://www.africamuseum.be/en/research/discover/news/scientificillustrations>
 - Because cameras are weird too, taking good photographs of a wet specimen etc
 - The illustration gives a clearer, more 'beautiful', but above all more accurate depiction of the specimen being described."

2019 example of a grad student deciding to use stippling for scientific illustration rather than photo?

- Current scientist still stippling for his work rather than photographs;

s a graduate student at the University of Michigan, I had a work-study job illustrating fossil fish bones for Gerald R. Smith, the Curator of Fishes (at the time). I illustrated those small fossils the traditional way, with a bottle of ink, a crow-quill pen, and a microscope with a camera lucida. Here's one example:
- Well... we do not photograph the invisible or the nano small until
-

And finally we reach the scanning electron microscope where each image produces is a pixel value sent digitally from the microscope

- Expand the SEM connection: The final sentence connecting SEM to stippling is a good thought, but needs more detail. An SEM doesn't literally "stipple" in the same way a pen does. Instead, it uses a scanning electron beam to gather pixel-by-pixel signal data. The image is a digital raster of brightness values based on electron interactions, which can produce a granular, dot-like appearance, but it's a parallel, not a direct continuation. A good clarification would be to explain that the underlying principle of building an image from discrete points of information is the powerful through-line, not the specific physical technique.
-
- n Wandelaar (1690–1759)
 - As an artist and engraver working with anatomist Bernhard Siegfried Albinus in the mid-18th century, Wandelaar is a prominent example of stippling's use in medical illustration.
 -

<https://www.hsm.ox.ac.uk/engraving>

Ultrasounding the ocean exhibit at the WHOI discovery center”
illustrates how scientists use sound waves like an ultrasound to
visualize underwater life

- Using interactive art to display a scientific process to public

Glass Flowers at the Natural Museum of History

- Educational material accessibility to students
- Magnification, accurate models

aranowsky_DeadFish_2016.pdf

Quotes:

- Specifically, they point to a tension inherent to, but rarely explicitly addressed, in the practice of field illustration: At what point in an animal's transition from life to death should an artist represent it? At what point is representation even possible?
 - Not a quote: This makes me think of when we visited the fish collection in the museum of natural history where all of them looked somewhat the same color, but the drawings were so beautiful and vivid
- axidermy, dioramas, or illustrations published in scientific atlases, for example—are often cast as depictions of enduring liveliness: they involve the taking of dead specimens and making them appear as if alive, forever suspended in what Donna Haraway famously described as a “moment of supreme life.”
- different approach: they document the immediate—if ephemeral—way the specimen appeared to him in the field. Rather than attempt to reanimate, Drayton thematized death and decay, making them central elements to the images at hand
- animal's former self, a version explicitly not depicted visually on the page, Drayton reveals an aspiration to capture a kind of artistic subject he knows to be impossible: a living, swimming fish. This is a version of the animal accessible only in fleeting impressions and partial glimpses, mediated by the distortive effects of light refracting in water, the fish's flittering body, and the dynamic play of shadow and light in a swiftly churning river environment. Rather than suppress all trace of himself, Drayton, in pursuit of this elusive subject, becomes a presence in the picture

Daston and Galison

Images, graphical representations curves, forms shapes - “the language of the phenomena itself”

“Let nature speak for itself” - scientific objectivity

The paper says there was a movement to remove human imposition of hope desire aesthetic generalization or expectation

- I would argue that mechanically produced images, were created of aesthetics... and continue to be
 - Stippling jump???
- Problem for atlas makers was not world vs mind, it was struggling with inward temptation
 - Not to fix things up
 - Eliminating the mediating presence of the observer was a problem for these people
 - Machines offer freedom from will

- Atlas's were meant to display something more objective than not

Some good quotes in there

"The atlas aims to make nature safe for science; to replace raw experience-the accidental, contingent experience of specific individual objects - with digested experience." pg 85

"Atlases supply working objects to the sciences of the eye."

"Atlases drill the eye of the beginner and refresh the eye of the old hand"

"Whatever the amount and avowed function of the text in an atlas, which varies from long and essential to nonexistent and despised, the illustrations command center stage... they are the *raison d'être* of the atlases"

"Simply pickling the anomalous organ in alcohol is a poor substitute for a faithful drawing of a fresh cadaver, since changes of form and color "denature them" and in any case such a specimen "profits only a small number of observers. Lacking the repetitive experience of normal anatomy, the pathologist is doubly aware of the psychological truth that a picture "constantly reproduces the same image" created a vivid, indelible memory"

"Truth to nature"

"For our purposes [Hunter's] naturalism is intrusive because it shows, first, that scientific naturalism and the cult of individuation detail long antedated the technology of the photograph; and, second, that naturalism need not be coupled with the anxiety of distortion and the rejection of aesthetics.

"Eighteenth century atlases required more than mere accuracy"

Characteristic images

Photographic depiction as - never wholly successful

- I think that Daston and Galison are saying that at the time the shift to mechanistic techniques tried to eliminate subjectivity etc but really, mechanistic techniques are also artistic, or at least based off of artistic principles just as a photograph does not tell the whole story of our perception of nature, it cannot fully encapsulate it

"Fwaigh closer to th represent, ANderson creditor art with an unprecedented importance, but he did so only by redefining art from the fine arts of the past into any form of schematic illustration, diagram, photograph, or model. In this new sense, artistic aids could "serve as a new language that speak with strength and clearness where written or spoken words could convey their meaning slowly and imperfectly"

- Context, talking about the shift to photography and mechanistic techniques

"Artists, even slavishly realistic ones, agreed that the artists' very presence meant that images were mediated. "

"Composite portraits"

Art in science attempted to "remove the process of abstraction from the artist's pen"

- Move towards pictorial objectivism, against subtler alterations etc

"Still it was clear that there were systematic mismatches between macroscopic anatomy and the x-ray image of the human body. There were elements of the body that did not produce image traces on the x-ray and there were representational elements on the x-ray that did not correspond to identifiable characteristics under the anatomist's knife, "

"The most a picture could do was to serve as a signpost"

"Burden of representation fell on the audience, instead of on the picture itself" (this begs education I feel like)

"Our belief guarantees its authenticity "

X-rays: "here the encryption of information takes place in the technology itself,.... Everybody knows that a skiagraph may be easily made to tell untruths and that a like unstable veracity may attach to skiagrams... photographs did not carry a transparent meaning. "

"The photograph did not end the debate over objectivity, it entered the debate. "

“Accuracy had to be sacrificed on the altar of objectivity “

“Pictures became more than merely helpful tools, they were the words of nature itself”

“There were many cases where the photograph was unable to provide the audience with a guide equal to that offered by an illustrator.

“Teach the reader how to recognize the typical

“Choulant praised Vesalius not only for his scientific methods but also for his “eye artistically trained” which led him to anatomical truth by “pursuing beauty in all the workings of nature: “

“The genial image fit best when art and science self-consciously converged, as in the case of the idealizing anatomist and botanists.”

MICHAEL LYNCH

In the abstract: movement away from theory centric view of science towards an interest in practical aspects - which is why visual representation has become more important

Scientific realism is no less nuanced than artistic realism, and, in fact, scientists have often made use of artistic conventions and techniques for documenting observations and illustrating texts (Edgerton 1975; Alpers 1983).

They may evoke aesthetic appreciation, when they are used strictly as devices for identifying birds in the field they are not likely to be judged by the same criteria as original works of art. Instead, they will be evaluated in terms of how well they enhance the birdwatcher's pragmatic activity (Law and Lynch 1990). This is not to argue for a rigid separation between aesthetic and pragmatic considerations. To the contrary, it can be argued that "aesthetic" criteria for judging illustrations are always bound-up with pragmatic contexts (Lynch and Edgerton 1988). The 'high' aesthetic of the art museum is not the only source of conventions and standards for crafting illustrations and assessing their visual qualities.

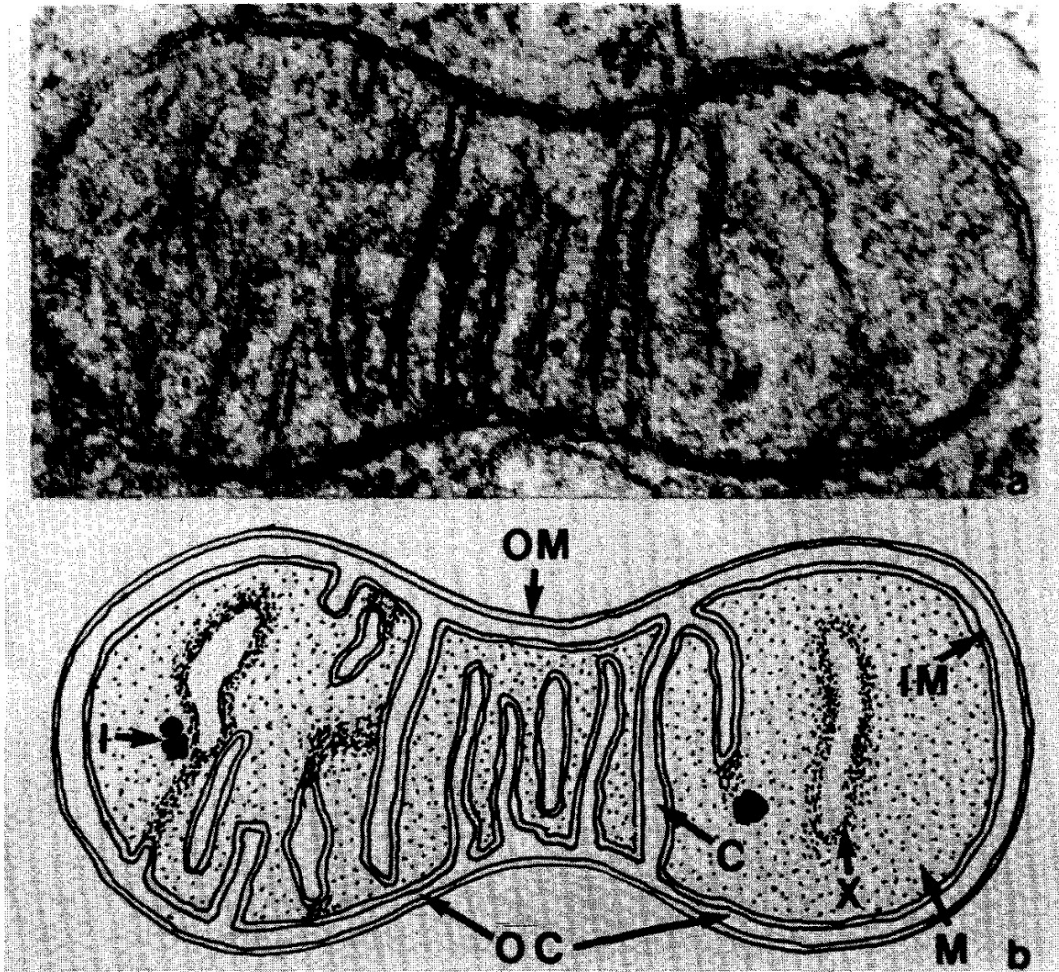
The artist's sleight of hand, of course, can also have indispensable advances, since it creates and embodies a concept-laden vision of things. When supported by a context of belief, the 'low' art of manual representation can be transformed into the 'high' science of conceptual understanding. Because the diagram provides a continuous field in which no clear demarcation is made between

conceptually assumed and concretely seen detail, it is often regarded as a suspect form of original evidence.

Something i thought of - Our eyes have color constancy - we can recognize the color and shape of an object or identify something regardless of the illumination

Klein Paper Tools Ursula Klein: (maybe use to qualify pig atlas?)

- Visibility and maneuverability are aspects of sign systems that they share with material objects, and more specifically with material tools used to achieve certain goals. We may designate these features their “material” dimension. Comparable to the materiality of ordinary tools, they define objective possibilities and constraints of work with a given type of sign system. Paper tools are material devices in the broader sense of being visible and maneuverable entities that are exterior to mental processes but help to generate mental processes.
- My comparison of constructive-additive representations of numbers in ancient Egypt with the use of Berzelian chemical formulae in nineteenth-century Europe has highlighted relatively simple forms of materiality, action, and thinking, which yield representations that can be directly related to real objects (“first-order representations”). The construction of such simple representations is accompanied by abstractions from local features and contexts of actions with objects. It thus yields insights into some general features of the objects under investigation. However, such kinds of (first-order) external representations are not yet mathematical concepts or scientific theories. If you want to know more about these issues, Peter has many interesting ideas to offer as well.⁸



Art is analysis

The distance between drawing and mechanical reproduction enables the photograph to stand for the original object the diagram analyzes

Emails from teacher about my project

Dear Rhea,

I hope your weekend's been going great.

It was good meeting with you yesterday. Your proposed question -- Does innovation in science require the creation of new forms of art? -- is terrific. To recap our conversation on Saturday, tomorrow afternoon (around 4) we'll be hoping to see some movement on these research questions:

1. What kinds of case studies will you be looking at? Will you look at one case study or a few?
2. What kinds of resources (archival, published papers, etc.) were you considering using?
3. Are you interested in "art" per se, or aesthetics more broadly?
4. Do you have a particular span of time you'd want to look at? What are the advantages/disadvantages of casting a wide versus a narrow net?
5. Don't talk about dependence.... Instead talk about the relationship between art and science, some other relationship a back and forth. Dependence is hard to argue, there is clearly a relationship, but what is it... maybe

See you in the classroom tomorrow!

Best

MPR

PET SCANNING

- Art history about grids and pixels

Meaning for the feedback loop

= science names

- Matter of poignancy to something and saying that is what we should be looking at

Examples from biology

Dive into three examples of the loop and

ALL COMPILED LINKS

Links:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8002099/>
[https://www.sciencedirect.com/science/article/pii/S1434461024000610#:~:text=Similarly%2C%20the%20geneticist%20Richard%20Goldschmidt,%E2%80%9D%20\(Egerton%2C%202013\).](https://www.sciencedirect.com/science/article/pii/S1434461024000610#:~:text=Similarly%2C%20the%20geneticist%20Richard%20Goldschmidt,%E2%80%9D%20(Egerton%2C%202013).)
<https://www.artmeetsscience.co/art-forms-in-nature-ernst-haeckel/?srsltid=AfmBOorAUpHgv4z9w7LQTW1xCf1KoZ0GFHusDErfDpV2laan82MFHyv8>
<https://www.sciencedirect.com/science/article/pii/S0960982219314368>
<https://anatomypubs.onlinelibrary.wiley.com/doi/abs/10.1002/ar.1090970103#:~:text=Page%201,orientation%20of%20the%20prepared%20sections.>
foucault the order of things
atlas daston and galison
paper tools klein
<https://news.northeastern.edu/2019/11/13/a-cosmic-data-visualization-celebrates-science-journal-natures-150-year-anniversary/>
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https://www.artmeetsscience.co/art-forms-in-nature-ernst-haeckel/?srsltid=AfmBOoqx36qM2ZteA3cgLnHMOF_Yk-Bqiy2Aq_HeaPWYTOVGp37ti3e0
https://www.readandcobooks.co.uk/blog/art-forms-in-nature-2/?srsltid=AfmBOorNEMFf10Cwvqm5Cwb1n6c7NLgWjPKF0Sa6hid_wGivg3AVQu8h
<https://paleonerdish.wordpress.com/2014/08/29/haeckel-and-the-legacy-of-early-radiolarian-taxonomists/>
<https://www.cambridge.org/core/books/abs/cambridge-foucault-lexicon/biohistory/DC86126073FDE96300B5FB5262CCC2CB>
<https://www.drawforbiodiversity.com/en/dessin-scientifique>
<https://theconversation.com/seeing-what-the-naked-eye-cant-4-essential-reads-on-how-scientists-bring-the-microscopic-world-into-plain->

sight-211666#:~:text=Though%20hand%20drawings%20have%20since,microscope%20as%20a%20scientist%20would.
<https://darchive.mblwhoilibrary.org/search?query=leuckart>
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<https://www.smithsonianmag.com/smart-news/see-the-original-17th-century-drawings-of-the-microscopic-world-robert-hooke-discovered-3345281/>
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<https://www.artgallery.nsw.gov.au/collection/artists/martens-conrad/>
<https://maverickbeyond.com/2019/02/05/microscopy-in-art/>
<https://www.smithsonianmag.com/blogs/national-museum-of-natural-history/2020/04/15/why-science-needs-art/>
<https://www.forbes.com/councils/forbestechcouncil/2023/05/12/the-art-of-innovation-how-scientific-research-and-the-arts-can-collaborate-to-generate-innovations/>
<https://direct.mit.edu/leon/article-abstract/43/1/59/45394/Similarities-and-Contrasts-in-the-Creative?redirectedFrom=fulltext>
<https://theconversation.com/seeing-what-the-naked-eye-cant-4-essential-reads-on-how-scientists-bring-the-microscopic-world-into-plain-sight-211666#:~:text=Though%20hand%20drawings%20have%20since,microscope%20as%20a%20scientist%20would.>
<https://theconversation.com/art-illuminates-the-beauty-of-science-and-could-inspire-the-next-generation-of-scientists-young-and-old-168925>
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#:~:text=Webb's%20new%20view%20of%20the,still%20forming%20come%20into%20view.
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<https://annemarieatkinson.co.uk/what-is-sciart/>
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<https://www.computerhistory.org/revolution/computer-graphics-music-and-art/15/228>
<https://unclespld3r.github.io/posts/2023/03/computer-history-the-evolution-of-computer-graphics/>
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